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Lecture 1 (DY) Tutorial on categorical representations

V $\mathbb{k}$ dim vec. sp. $f: V \rightarrow V$ endo.

$$k \rightarrow V \otimes V^* \xrightarrow{f \otimes \text{id}} V \otimes V^* \rightarrow k \quad \text{determines } \text{Tr}(f, V) \in k.$$

Let $C \in \text{Ducat}$, $F \in C$ (all morphisms of Ducat are colimit preserving)

If C is dualisable. $\text{Vect} \rightarrow C \otimes C^V \xrightarrow{F \otimes \text{id}} C \otimes C^V \rightarrow \text{Vect}$

$$\underbrace{\hspace{15em}}_{\otimes \text{Tr}(F, C) \in \text{Vect}}$$

Convention: for the rest of the talk, $k = \bar{k}$

1. $\text{char } k = 0$, $\text{Shv}(X) = \text{Dmod}(X)$, $\text{Cat} = \text{Ducat}$

For $X, Y \in \text{Var}$, $D(X) \otimes D(Y) \simeq D(X \times Y)$

For $f: X \rightarrow Y$, get functors $f_*: D(X) \rightarrow D(Y)$

$f^!: D(Y) \rightarrow D(X)$

2. $\text{char } k \neq 0$, $\text{Shv}(X) := \underline{\text{Shv}}(X)$ ℓ -adic constr. sheaves on X , $\text{Cat} = \text{AlgCat}$

$\underline{\text{Shv}}(X) \otimes \underline{\text{Shv}}(Y) \simeq \underline{\text{Shv}}(X \times Y)$

$f: X \rightarrow Y$, get f_* , $f^!$.

In case 2: $\text{Shv}(X) \ni \text{Fr}$

$$\text{Tr}(\text{Fr}, \text{Shv}(X)) = \overline{\mathbb{Q}}_\ell[X(\mathbb{F}_q)] \quad \begin{array}{l} \text{pull back to } \Delta \\ \text{push to a pt} \end{array}$$

$$\text{Vect} \rightarrow \text{Shv}(X \times X) \xrightarrow{F \times \text{id}} \text{Shv}(X \times X) \rightarrow \text{Vect}$$

$\Delta_* 1_X \quad (\text{id}, F)_* 1_X \quad \Delta \cap (\text{id}, F) = X(\mathbb{F}_q)$

$C, D \in \text{Cat}$

$$C \xrightarrow{a} D$$

assume a has a right adjoint
 $\in \text{Cat}$

$$\begin{array}{ccc} F_C & \nearrow & \downarrow F_D \\ C & \xrightarrow{a} & D \end{array}$$

gives map $\text{Tr}(a): \text{Tr}(C, F_C) \rightarrow \text{Tr}(D, F_D)$

Motivation: G reductive gp / $(\mathbb{F}_q((t)))$, LLC classifies ^{smooth} rep's of $G(\mathbb{F}_q((t)))$

i.e. rep's of $H(G(\mathbb{F}_q((t))))$ Hecke algebra

$\text{Shv}(L_G)$

write $L_G \simeq \text{colim } L_{G_{\leq \lambda}}$

$\uparrow$
ind-scheme

$\uparrow$
placid schemes

$L_{G_{\leq \lambda}} \simeq \lim \frac{L_{G_{\leq \lambda}}}{G_{\text{norm}}}$

finite type scheme

$$\text{Tr}(F, \text{Shv}(L_{G_{\leq \lambda}})) \simeq \text{Tr}(F, \text{colim } \text{Shv}(L_{G_{\leq \lambda}}/G_{\text{norm}}))$$

$$\simeq \text{colim } \overline{\mathcal{O}}_G[L_{G_{\leq \lambda}}/G_{\text{norm}}] \simeq \overline{\mathcal{O}}_G[L_{G_{\leq \lambda}}]_{\text{loc. const.}}$$

$$\text{Tr}(F, \text{Shv}(L_G)) \simeq \text{Tr}(F, \text{colim } \text{Shv}(L_{G_{\leq \lambda}})) \simeq \text{colim } \overline{\mathcal{O}}_G[L_{G_{\leq \lambda}}]_{\text{loc. const.}} \simeq H.$$

$$\text{Tr}(F, \text{Shv}(L_G)) \simeq H$$

modules for $\text{Shv}(L_G)$ category H -rep's

Local geom. Langlands (Conjecture)

$$\underline{\text{Shv}(L_G)\text{-mod}} \simeq \underline{\text{Cat}/\text{Loc Sys}_G}$$

For a 2-cat. $\mathcal{C} \hookrightarrow F$, also can define $\text{Tr}(F, \mathcal{C})$

$$\begin{array}{ccc} & \text{Shv}(G)\text{-mod}^{(\text{res})} & \\ & \text{Tr}(\text{id}_1, -) \searrow & \swarrow \text{Tr}(F, -) \\ (CS(G)) & \text{Shv}(G/G) & \overline{\mathcal{O}}_G[G(\mathbb{F}_q)]\text{-rep} \\ & \text{Tr}(F, -) \searrow & \swarrow \text{Tr}(\text{id}_1, -) \\ & \overline{\mathcal{O}}_G[G/G] & \text{Page 2} \end{array}$$

For a point $x \in B(G)$, $r \in \mathbb{R}_{\geq 0}$. G gp over $k((t))$

$$K_{x,r+t}, K_{x,r} \subset L_G \quad \text{For } C \subset \text{Shv}(L_G),$$

want to define $C^{\leq r}$ to be the L_G -subset of C gen'd by $G_{x,r+t}$ -invariants.

$$C^{G_{x,r+t}} \simeq \text{Hom}_{\text{Shv}(G_{x,r+t})}(\text{Vect}, C)$$

Lemma (Ben-Zvi - Lunningham - Viehweg)

For any $C \subset L_G$

$$\text{Shv}(L_G / G_{x,r+t}) \otimes_{\text{Shv}(K_{x,r+t} / L_G / G_{x,r+t})} C^{G_{x,r+t}} \longrightarrow C$$

fully faithful,

& has L_G -equiv. right adjoint.

Can define $C^{\leq r}$ to be the smallest subset of C containing the images of all these functors.



$$C^{K_{x,r+t}} \hookrightarrow \text{Shv}(K_{x,r} / K_{x,r+t}) \quad \text{because } K_{x,r+t} \triangleleft K_{x,r}$$

is

$$\text{Shv}((K_{x,r} / K_{x,r+t})^*)$$

$$(C^{\leq r})^{K_{x,r+t}} \simeq C^{K_{x,r+t}}$$

Th (Moy - Prasad)

$$P_x / K_{x,0+t} \simeq (K_{x,r} / K_{x,r+t})^*$$

$$(K_{x,r} / K_{x,r+t})^{*,us} = \left\{ p \in (K_{x,r} / K_{x,r+t})^* \text{ whose } P_x / K_{x,0+t} \text{-orbit contains } 0 \text{ in its closure} \right\}$$

$$(C^{<r})^{K_{x,r+}} \simeq C^{K_{x,r+}} \otimes_{\text{Shv}((K_{x,r}/K_{x,r+})^*)} \text{Shv}((K_{x,r}/K_{x,r+})^{*,us})$$

If $(K_{x,r}/K_{x,r+})^{*,o}$ is the complement of $(K_{x,r}/K_{x,r+})^{*,us}$,

then ~~*~~ $C^{<r} \simeq \phi$, ~~iff~~ $C^{K_{x,r+}} \otimes \text{Shv}((K_{x,r}/K_{x,r+})^*)$

factors through $\text{Shv}((K_{x,r}/K_{x,r+})^{*,o})$

for all x .

Lecture 2 (YV) G unimodular reductive gp / $F = k((t))$

$$L_G\text{-cat} = \text{Shv}(L_G)\text{-mod}, \quad B(G) = \text{reduced building}$$

$$\forall x \in B(G), \quad L_G \supset \underset{\text{stab}_{L_G}(x)}{G_x} \supset G_{x,o} = G_x^o \supset G_{x,r} \supset G_{x,r+}, \quad r \in \mathbb{Q}_{>0}.$$

$$L_x = G_{x,o} / G_{x,o+}$$

$$\forall C \in L_G\text{-cat}, x \in B(G), r \geq 0, C^{G_{x,r+}} \in \text{cat}$$

$$\text{Let } C^{\leq r} = \left\{ \text{smallest } L_G\text{-inv. subcat of } C \text{ containing } C^{G_{x,r+}}, \forall x \in B(G) \right\}$$

$$C^{<r} = \bigcup_{s < r} C^{\leq s}$$

$$L_G\text{-cat}^{\leq r} = \{ C \in L_G\text{-cat} : C^{\leq r} = C \}, \quad L_G\text{-cat}^{=r} = \{ C \in L_G\text{-cat}^{\leq r} : C^{<r} = 0 \}$$

Th (Mazhar-Masad, Yang) $L_G\text{-cat}$ has SOD by $L_G\text{-cat}^{=r}$.

$$\forall x \in B(G), C^{G_{x,r+}} \in G_x / G_{x,r+}\text{-cat}$$

$$G_x / G_{x,r+} \supset G_{x,r} / G_{x,r+} \xrightarrow[\text{mp isom pages}]{\simeq} G_{x,r} / G_{x,r}$$

$\Rightarrow C^{G_{X,r+}}$ is module cat. w.r.t.

$$\left(\text{Shv}(g_{X,r}/g_{X,r+}), * \right) \underset{FT}{\simeq} \left(\text{Shv}((g_{X,r}/g_{X,r+})^*), \overset{!}{\otimes} \right)$$

$$\text{Let } (g_{X,r}/g_{X,r+})^{*,0} = \left\{ v \in (g_{X,r}/g_{X,r+})^* : \overline{L_X \cdot v} \neq 0 \right\}$$

Th (MP, Yang)

$C \in \text{LH-cat}^{S_r}$ belongs to $\text{LH-cat}^{=r}$ iff the action of $\text{Shv}((g_{X,r}/g_{X,r+})^*, \overset{!}{\otimes})$

on $C^{G_{X,r+}}$ factors through $\text{Shv}((g_{X,r}/g_{X,r+})^{*,0})$, $\forall x \in B(G)$



II. $M \subset G$ is finite Levi iff $M = Z_G(S)$, $S \subset G$ is a torus

$$\Leftrightarrow M = Z_G(x), x \in g_{SS}.$$

Goal. $\text{LH-cat}^{=r}$ has SOD

$$\left\{ \text{LH-cat}^{=r, M} \right\}_{M \in \text{TL}(G)} \rightleftharpoons \left\{ \begin{matrix} M \subset G \\ \text{tL} \end{matrix} \right\} / \text{conj.}$$

$$\text{Step 1 } \forall x \in B(G), \exists \text{ stratification } (g_{X,r}/g_{X,r+})^{*,0} = \coprod_{M \in \text{TL}(G)} (g_{X,r}/g_{X,r+})^{*,M}$$

$$\text{Step 2 } \forall x \in B(G), \text{ let } G_x/G_{X,r+} - \text{cat}^M = \left\{ C \in G_x/G_{X,r+} - \text{cat} : \right.$$

the action of $\text{Shv}((g_{X,r}/g_{X,r+})^*)$ on C factors through

$$\text{Shv}((g_{X,r}/g_{X,r+})^*) \xrightarrow{\overset{!}{\text{in}}} \text{Shv}((g_{X,r}/g_{X,r+})^{*,M}) \}$$

i.e. $\forall M' \neq M, F \in \text{Shv}((g_{x,r}/g_{x,r+})^{*,M'})$ acts by zero.

Step 3. $L_{\mathcal{G}}(\text{cat}^{\text{tr}}, M) = \left\{ C \in L_{\mathcal{G}}(\text{cat}^{\text{tr}} : \forall x \in B(\mathcal{G}), C^{g_{x,r+}} \in g_x / g_{x,r+} - \text{cat}^{\text{tr}} \right\}$

Th. $\left\{ L_{\mathcal{G}}(\text{cat}^{\text{tr}}, M) \right\}_M$ is a SOD of $L_{\mathcal{G}}(\text{cat}^{\text{tr}})$.

Goal $\forall \xi \in (g_{x,r}/g_{x,r+})^* \rightsquigarrow \tilde{\xi} \in L(g(k) = g(F))$

$$\rightsquigarrow \tilde{\xi}_{ss} \in g(F) \rightsquigarrow M = Z_{\mathcal{G}}(\tilde{\xi}_{ss})$$

Assume for simplicity $\text{char } k \gg 0$. (we need $\mathcal{G}$ is tame + char $k \nmid |W|$)

$$\text{s.t. } k: g \xrightarrow{\sim} g^*$$

Then $k: (g_{x,r}/g_{x,r+})^* \xrightarrow{\sim} g_{x,-r}/g_{x,-r+}$

Claim $g_{x,r+} \rightarrow g_{x,-r}/g_{x,-r+}$ has a "natural" section.

Assume $\mathcal{G}$ split $\Rightarrow$ then $g_{x,-r} = \bigoplus_{\langle x,d \rangle + n \geq -r} t^n g_d$

$$g_{x,-r+} = \bigoplus_{\langle x,d \rangle + n > -r} t^n g_d$$

$$\text{So } g_{x,-r}/g_{x,-r+} \simeq \bigoplus_{\langle x,d \rangle + n = -r} t^n g_d \subset g_{x,-r}$$

Rem 1 Section depends on a choice of $t \in K((t))$.

Rem 2 $Z_{\mathcal{G}}(\tilde{\xi}_{ss})$ does not. Moreover, $M_{\tilde{\xi}}$ is the largest centralizer among all sections $\tilde{\xi} \mapsto \tilde{\xi}_{ss}$

Def $(g_{x,r}/g_{x,r+})^{*,M} = \{ \xi : M_{\tilde{\xi}} = M \}$.

Main result

Variant: Let $M \subset G$ twisted Levi.

$$\left(m_{x,r} / m_{x,r+} \right)^{*,M} > \left(m_{x,r} / m_{x,r+} \right)^{*,M, \text{gen}/G}$$

$$= \left\{ \tilde{\gamma} : \sum_G(\tilde{\gamma}_{ss}) = M \right\}$$

(not only $\sum_M(\tilde{\gamma}_{ss}) = M$)

$$\rightsquigarrow LM\text{-cat} = r, M, \text{gen}/G$$

Main Th. Assume G -tame, char $k \nmid |W|$, $\forall r > 0$, $M \in TL(G)$, $\exists$ equiv of $(\infty, 2)$ -cat,

$$LG\text{-cat} = r, M \rightsquigarrow \left(LM\text{-cat} = r, M, \text{gen}/G \right)^{W_{G,M}} \quad W_{G,M} = N_G(M)/M.$$

Proof Let $G_{ab} = G/[G, G]$, $G \rightarrow G_{ab}$. Then $\exists$ equiv. of 2-cat.

$$LM\text{-cat} = r, M \hookrightarrow L(M_{ab})\text{-cat} = r \quad \otimes \quad LM\text{-cat} < r$$

$$L(M_{ab})\text{-cat} < r$$

In other words, $\forall C \in LM\text{-cat} = r, M$ is of the form $\pi \otimes C'$ s.t.

$$C' \in LM\text{-cat} < r, \quad \pi \in L(M_{ab})\text{-cat}.$$

Lecture 3 (GP) Cartoon. $\pi_\lambda, \lambda \in \eta^*$ typically breakup η^* into pieces s.t.

rep theory of G in a given piece $\sim$ rep theory of $M \hookrightarrow G$ w/ some parameters

Version today. $LG \simeq \mathcal{E}$, depth = r

$$x \in B(G)$$

$$e^{G_{x,r+}}$$

$$\downarrow$$

$$g_{x=r}^*/G_x$$

$\mathcal{P}$

(up to rel. Weyl gp)

ex. principal series use pind & pres

$$g = u^- \oplus m \oplus u$$

Th (DLV) $M \in \text{TL}(G)$,

$$L_G\text{-cat}^{=r, M} \simeq \left(L_M\text{-cat}^{=r, M\text{-gen}} \right)^{W_{M, G}}$$

Today: sketch of pr.

Strategy: 1. reduce to a statement point by point in $B(G)$ (localisation th).
2. prove that statement

Localization

$$L_G \simeq e \rightsquigarrow \left\{ e^{G_{x, r+}} \right\}_{x \in B(G)}$$

structures: 1. $G_x \simeq e^{G_{x, r+}}$

2. if $x, y \in B(G)$, $G_{x, r+} \subset G_{y, r+}$, then

$$e^{G_{y, r+}} = \left(e^{G_{x, r+}} \right)^{G_{y, r+}}$$

3. L_G -equiv:

$$g \cdot - : e^{G_{x, r+}} \simeq e^{g \cdot G_{x, r+}}$$

Upshot: $L_G\text{-cat}^{sr} \rightarrow \text{"ShiCat"}^{L_G}(B(G))$

Th is an equivalence. $e \rightarrow \varprojlim_{\substack{A_v \\ x \in B(G)}} e^{G_{x, r+}}$

Rem. 1. Alu's - Curtis duality, Deligne - Lusztig's duality, Aubert - Zelevinsky duality
Schneider - Stuhler Lec. for p -adic groups

idea $L(\mathbb{F}_q) \quad C_*(B(G)) = \begin{cases} * = 0 & \text{triv} \\ * = \text{top} & \text{st} \end{cases}$

$L(\mathbb{Q}_p) \quad C_*(B(G)) = \begin{cases} * = 0 & \text{triv} \\ 0 & \text{o.w.} \end{cases}$

2. (Schneider - Stuhler) $C_x(B(G), \{ \pi^{G_{x, r+}} \}) \xrightarrow{\sim} \pi$

2. Passing to a Levi at $x \in B(G)$.

$$M \hookrightarrow G \text{ TL. } , x \in B(M) \hookrightarrow B(G)$$

$$N_x = N_G(M) \cap h_x$$

Th. $\exists$ canonical equiv.

$$h_x^* / h_{x, r+} - \text{Cat}^M \xrightarrow{\sim} \bigoplus_{s \in S} N_x / M_{x, r+} - \text{Cat}^{M, \text{gen}}$$

Rem ~~Conclusion~~

S set of h_x -conj. classes of such M .

ex. $G = SL_2(\mathbb{C})$, $x = 0 \in \text{std apartment}$, $G_x = SL_2(\mathbb{C} \oplus \mathbb{C})$

in this case, $g_{x=r}^* = \begin{cases} 0 & \text{unless } r \in \mathbb{Z} > 0 \\ sl_2^* & , r \in \mathbb{Z} > 0 \end{cases}$

stratification is $sl_2^* = N \sqcup sl_2^{* \text{rss}}$ unstable locus, $sl_2^{* \text{rss}} / SL_2 \cong t^{* \text{rss}} / W$

In general, for $sl_2 \rightsquigarrow g$ split, $x = 0$, Lusztig's stratification.

$\subset \begin{matrix} g^{* \text{rss}} \\ \times \\ g^{* \text{rss}} \end{matrix} \times \begin{matrix} t^{* \text{rss}} \\ \times \\ t^{* \text{rss}} \end{matrix}$ need N_x in a decomposition

In general, $g = m \oplus m^c$.

$$e \rightarrow \left(\left(e \times_{g_{x=r}^*} m_{x=r}^{* \text{M, gen}} \right)^{M > r/2} \right) \otimes_{H_{M^c=r/2}}^W W \quad \uparrow \text{universal wt-1}$$

V sympl v.s.

$$1 \rightarrow G_a \rightarrow H_V \rightarrow V \rightarrow 1 \quad \text{Heis}$$

χ : exp char sheaf on G_a

$$H_V - \text{Cat}^{\text{gen}} := H_V - \text{Cat} \text{ on which } G_a \text{ acts through } \chi$$

Stone-von Neumann: $H_V - \text{Cat}^{\text{gen}} \cong \text{Cat}$ Weil-Heis $\leftrightarrow$ Vect

$$V = L \oplus L^*, \quad Sh(L) \cong \text{Weil-Heis}$$

$$\hookrightarrow Sh(Sp \ltimes H_V)$$

Moreover, nat'l "projective" action of $Sp(V)$ on gen is non-obstructed

Lecture 4 (DY)

LG-cat built out of $G_x/G_{x,r+} - \text{mod}$

$$\mathbb{I}_{x,r} : G_x/G_{x,r+} - \text{mod}^{=r,M} \simeq N_x/M_{x,r+} - \text{mod}^{=r,M, \text{gen}}$$

$$\ell \mapsto \text{Hom}(WH, e^{G_x, \frac{r}{2}+}/G_{x,r+} M_{x, \frac{r}{2}+})$$

From LG-cat $\rightsquigarrow$ system of categories $x \rightsquigarrow C_x, x \in B(G)$

$$K_{x,r+} \subset K_{y,r+} \subset K_{y,r} \subset K_{x,r}$$

$$C_x^{K_{y,r+}/K_{x,r+}} \simeq C_y$$

want $\mathbb{I}_{x,r}(C_x) \quad x \in B(M)$ to be a system of cat on $B(M)$

$\rightarrow$ produce M-cat colim $\mathbb{I}_{x,r}(C_x)$

$$\text{NOT TRUE: } \mathbb{I}_{x,r}(C_x)^{M_{y,r+}/M_{x,r+}} \simeq \mathbb{I}_{y,r}(C_y)$$

true up to a quadratic char. $\rightarrow \otimes \left(\det_{G_{y, \frac{r}{2}}/G_{x, \frac{r}{2}+}} \cdot \det_{M_{y, \frac{r}{2}}/M_{x, \frac{r}{2}}}^{-1} \right)^* \text{sgn}$

V symplectic space, $M \rightarrow Sp(V)$

Assume $\exists$ L Lagrangian fixed by M Nature repⁿ of $H_V \rtimes M \curvearrowright \text{Shv}(L)$

This repⁿ does NOT extend to $H_V \rtimes Sp(V)$.

$$WH|_{H_V \rtimes M} \simeq \text{Shv}(L) \otimes \underbrace{\text{quad. char.}}_{\text{Vect}_{\det^* L} \text{sgn}}$$

Rem Let A be a finite gp. Character of A $\Leftrightarrow$ action of A $\curvearrowright$ k

If G is an alg gp, mult. char. sheaf of G $\Leftrightarrow$ action of G $\curvearrowright$ Vect

sgn is the nontrivial order 2 char. sheaf on G_m

Imagine for all $x \in B(M)$, had a char. sheaf ψ_x on N^x

s.t. $\psi_x|_{N_y} \simeq \psi_y \otimes \text{discrepancy}$

$y \in \bar{x}$

Then $x \mapsto \mathbb{I}_{x,r}(C_x) \otimes \text{Vect } \psi_x$ would be a system of cat / $B(M)$.

Can do a bit weaker: Each $M_x/M_{x,r+}$ - cat. lives over $(\mathbb{Z}_m)_r/(\mathbb{Z}_m)_{r+})^*$

so suffices to find a family of char. sheaves over $(\mathbb{Z}_m)_r/(\mathbb{Z}_m)_{r+})^* \supset W_M = N(M)/M$

Rem. 1. ext'n of FKS char from M^x to N^x .

2. This ext'n is unique but not up to unique isom. (responsible (probably) for various $\mathbb{Z}/2$)
3. Can construct other systems of characters satisfying same property.

Proven $G\text{-cat}^{=r,M} \simeq (M\text{-cat}^{=r,M,\text{gen}})^{W_{\text{rel}}}$

$$M\text{-cat}^{=r,M,\text{gen}} \simeq M\text{-cat}^{<r} \otimes \text{Sh}_v\left(\left((\mathbb{Z}_m)_r/(\mathbb{Z}_m)_{r+}\right)^*\right)\text{-mod}$$

To define $G\text{-cat}^{\text{res}}$: $C \in G\text{-cat}^{\text{res}}$

1. C is res. $\Leftrightarrow$ each $C^{=r,M}$ is res.

$\Downarrow$

$$C \simeq \bigoplus C^{=r,M}$$

2. $C \in G\text{-cat}^{=r,M}$, C is res. $\Leftrightarrow \mathbb{I}(C)$ is res.

3. $C \in M\text{-cat}^{=r,M}$, C is res. $\Leftrightarrow C_1, C_2$ res.

$$\begin{array}{ccc} & \text{is} & \\ & C_1 \otimes C_2 & \\ \uparrow & & \uparrow \\ M\text{-cat}^{<r} & \text{Sh}_v(-)\text{-mod} & \end{array}$$

4. $C \in G\text{-cat}^{=0}$

$$G\text{-cat}^{=0} \simeq \text{Sh}_v\left(I^+ \backslash LG/I^+\right)\text{-mod}$$

$$\begin{array}{ccc} \cup & & \uparrow \\ G\text{-cat}^{=0,\text{res}} & \simeq & \text{Sh}_{v_{\text{mon}}}\left(I^+ \backslash LG/I^+\right)\text{-mod} \end{array}$$

Define $D(\text{CS}^{\text{att}}(G)) \cong \text{Tr}(\text{id}, \text{LG-Cat}^{\text{res}})$

Rem. restricted LGL

$$\text{Shv}(\text{LG})\text{-cat}^{\text{res}} \simeq \text{Cat} / \text{LocSys}_{\check{G}}^{\text{res}}$$

For each semisimple $\sigma \in \text{LocSys}$,

$(\text{LocSys}_{\check{G}})_{\sigma}$ = local systems w σ semisimplification,

$$\text{LocSys}_{\check{G}}^{\text{res}} = \coprod_{\sigma} \text{LocSys}_{\check{G}}^{\sigma}$$

(Main Theorem $\Rightarrow$ restricted LGL)

+ many things

Bezrukavnikov:

$$\text{Shv}(I \backslash \text{LG} / I) \simeq \text{Indcolim}_{\text{unip}} \left((\tilde{N} \times_{\check{G}} \tilde{N}) / \check{G} \right)$$

$$\Rightarrow \text{LG-Cat}^{\text{I-generated}} \simeq \text{Cat} / \text{LocSys}_{\check{G}}^{\text{tame, unip}} \quad (\text{Dhillon - Li - Yun - Zhu})$$

generalises depth 0 $\text{Shv}(I^+ \backslash \text{LG} / I^+)^{\text{res}}$

$$\Rightarrow \text{LG-Cat}^{=0, \text{res}} \simeq \text{Cat} / \text{LocSys}_{\check{G}}^{=0, \text{res}}$$

$\text{Tr}(\text{Fr}, \text{restricted LGL})$ = categorical LLC

Induct: assume we know RLGL for depth $< r$.

For depth r , $\text{LG-Cat}^{=r, M, \text{res}}$

$$\cong \text{colim}(\text{LocSys}_{\check{G}}^{\text{res}})$$

$$(\text{M-cat}^{=r, M, \text{gen, res}})^{W_{\text{rel}}}$$

(1)

$$(\text{M-cat}^{<r} \otimes \text{Shv}(\text{)-cat})^{W_{\text{rel}}} \simeq$$

$$\text{Cat} / \text{LocSys}_{\check{M}}^{=r, M, \text{res}}$$

(2) [Dat - Helm - Kirschenik

$$\text{Cat} / \text{LocSys}_{\check{M}}^{=r, M, \text{gen, res}} / W_{\text{rel}}^{\text{arithm}} \quad \text{[Moss in arithmetic]}$$

(3)

$$\text{Cat} / (\text{LocSys}_{\check{M}}^{<r} \times \dots) / W_{\text{rel}}$$

"explicit = fusion"